

Research

Characterization of novel peptides with antimicrobial and immunomodulatory potential

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Abstract

In the present study, we designed six peptides based on key characteristics of natural antimicrobial peptides (AMPs), including small size, cationicity, amphipathicity, and α -helical structure. The peptides were evaluated for their antimicrobial activity, hemolytic potential, binding affinity to TLR4/MD-2 receptors, and immunomodulatory effects. Using the AxPEP server, five of the six peptides were classified as AMPs, while one was identified as a non-antimicrobial peptide (NAMP). The antimicrobial activity of the designed peptides varied against *Escherichia coli* (*E.coli*), *Staphylococcus aureus* (*S.aureus*), *Klebsiella pneumoniae* (*K.pneumoniae*), and *Pseudomonas aeruginosa* (*P.aeruginosa*), with SK1217 and SK1260 being the most potent. Hemolytic assays revealed that SK1203, SK1217, SK1260, and SK1281 caused minimal red blood cell (RBC) lysis, indicating low hemolytic potential. In contrast, SK1286 and SK1283 displayed significant hemolysis at higher concentrations, with SK1286 inducing complete hemolysis at 100 $\mu\text{g/mL}$. Molecular docking analysis demonstrated strong binding of SK1260, SK1217, and SK1286 to TLR4/MD-2 receptors. Notably, SK1260 exhibited robust immunomodulatory activity, suppressing the release of pro-inflammatory cytokines IL-6 and TNF- α in LPS-activated macrophages. Cytotoxicity assays showed minimal toxicity for SK1260. Treatment with SK1260 enhanced the white blood cell count and increased cytokine levels (GM-CSF, INF- γ , and MCP-1) in the peritoneal lavage fluid of infected mice. Collectively, these findings underscore the potential of these cationic AMPs, particularly SK1260, as promising candidates for antimicrobial and immunomodulatory therapeutic applications.

Keywords Antimicrobial peptides · Cationic peptides · Hemolytic activity · TLR4/MD-2 receptors · Immunomodulatory cytokines · Leukocyte recruitment

1 Introduction

The rising resistance of microorganisms, including fungi, viruses, parasites, and bacteria such as staphylococci, enterococci, and *Escherichia coli*, to drugs has emerged as a critical global health challenge. Antimicrobial peptides (AMPs), produced naturally by most living organisms to ward off invasive pathogens, are excellent candidates for the development of novel antibiotics due to their inherent antimicrobial properties (R. [13, 30, 37]). Additionally, some AMPs

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have demonstrated spermicidal and anticancer attributes [2, 16]. These peptides are cationic and amphipathic and generally have a molecular weight under 10 kDa, showcasing diverse lengths, sequences, and structures. The first of these peptides was discovered in a giant silk moth, *Hyalophora cecropia*, in 1981 [27]. Since then, peptides with antibacterial properties have been identified across the spectrum of natural organisms, including bacteria, insects, amphibians, and mammals [4, 5, 10]. They typically exert their antimicrobial effect through three primary mechanisms: the carpet model, the barrel-stave model, and the toroidal-pore model. In the carpet model, AMPs align on the membrane surface like a carpet, accumulating until a critical concentration is reached, leading to membrane disintegration in a detergent-like manner [26]. The barrel-stave model involves AMPs inserting perpendicularly into the lipid bilayer, forming a transmembrane pore that disrupts ion balance and causes bacterial death [36]. In the toroidal-pore model, peptides induce lipid bending, forming dynamic pores where lipid head groups line the channel, maintaining membrane continuity while allowing leakage of cellular contents [15]. These diverse mechanisms highlight the versatility of AMPs in targeting bacterial membranes.

AMPs are categorized into five groups [24] (1) peptides forming alpha-helical structures, (2) peptides rich in cysteine residues; (3) peptides forming beta-sheet structures; (4) peptides rich in regular amino acids like histidine, arginine, and proline; and (5) peptides composed of uncommon and modified amino acids. These general characteristics guide the design and synthesis of potent AMPs [6, 14, 19].

This study aimed to improve the problems with current natural AMPs, mainly their potential harm to human cells, by creating new more effective peptides against microbes while being less harmful to human cells. The rationale for designing these peptides hinged on critical considerations such as charge distribution, hydrophobicity, and amino acid composition, all aimed at optimizing their antimicrobial potential specificity [1, 8, 14]).

Toll-like receptor 4 (TLR4), in association with myeloid differentiation factor 2 (MD-2), plays a crucial role in the innate immune response by recognizing lipopolysaccharides (LPS) from Gram-negative bacteria and triggering pro-inflammatory signaling cascades [23]. Given the immunomodulatory potential of antimicrobial peptides (AMPs), we selected the TLR4/MD-2 complex to investigate how our designed peptides interact with innate immune signaling pathways. Targeting this receptor is particularly relevant as modulation of TLR4 activity can help regulate inflammation, thereby offering therapeutic benefits for infections and inflammatory conditions [32]. By evaluating the binding affinity of our peptides to TLR4/MD-2, we aim to explore their potential role in immune modulation alongside their antimicrobial properties.

We synthesized a total of six peptides, drawing inspiration from the features of natural AMPs. We designed these peptides to be preferentially cationic, fostering better interactions with the negatively charged microbial membranes and thereby promoting efficient antimicrobial activity. We engineered these peptides to assume alpha-helical structures, which are known to potentiate membrane disruption and permeabilization, thereby elevating their efficacy in eliminating target microorganisms. We designed the peptides to securely anchor onto microbial membranes, aiding their insertion into the lipid bilayer and enhancing membrane destabilization, by incorporating over 30% hydrophobic residues. This optimization of hydrophobicity leads to improved antimicrobial potency and target specificity.

Our peptide design strategy introduces a novel approach that complements existing methodologies reported in the literature. Despite identifying numerous naturally occurring AMPs, their specific characteristics may not align with the precise optimization objectives pursued in this study. Our tailored peptides represent a novel combination of features that augment their antimicrobial effectiveness while minimizing the risk of cytotoxicity [6, 18].

In our study, one of the synthesized peptides, SK1260, showed great immunomodulatory activity in innate immune cells. It also had strong antibacterial effects without breaking down human blood red cells. The remarkable performance of peptide SK1260 positions it as a promising candidate for treating infectious diseases, reflecting the potential of our designed peptides to address the escalating challenges posed by drug-resistant microorganisms. By carefully adjusting the charge distribution, alpha-helical structure, and hydrophobicity, these peptides present an exciting way to develop new medicines that can fight antimicrobial resistance in the future.

2 Materials and methods

2.1 Design of peptide

The antibacterial peptides were designed using the APD (<http://aps.unmc.edu/AP/mainphp>) (G. [34], Z. [35]) and predicted using AxPEP server: <https://cbbio.online/AxPEP/>. The peptide sequences were chosen based on the following

Table 1 The sequence of the designed peptides

Peptide Id	Peptide sequence	Amino acid length
SK1203	AAWWKAKKWAWAKAWWKAW	21
SK1217	AFWWKAKKFAWAKKFAWWKAW	21
SK1260	KAFVVKFAWKFHAWKAWKKAW	21
SK1286	HFAPKSKPAHFPHKSKPKAW	21
SK1281	HFAVKHAVVAHVWHKAKSR	19
SK1283	HFAVKKAKVAHFVHKAKSKAW	21

Table 2 Bacterial strains and culture conditions

Bacterial strain	Gram stain	Species	Culture conditions
<i>E. coli</i> ATCC 8739	Gram-negative	<i>Escherichia coli</i>	Growth in LB broth at 37 °C
<i>S. aureus</i> ATCC 6538	Gram-positive	<i>Staphylococcus aureus</i>	Growth in nutrient agar at 37 °C
<i>K. pneumoniae</i> ATCC 700603	Gram-negative	<i>Klebsiella pneumoniae</i>	Growth in nutrient broth at 37 °C
<i>P. aeruginosa</i> ATCC 9027	Gram-negative	<i>Pseudomonas aeruginosa</i>	Growth in TSB medium at 37 °C

rules: (1) containing positive charged amino acids, (2) containing α -helix, and (3) containing hydrophobic amino acids. The sequences of six designed peptides are shown in Table 1.

2.2 The strains and growth conditions

One Gram-positive bacteria, and 3 Gram-negative bacteria, were selected to measure the antibacterial activity of peptides. The reference strains were provided by the Department of Microbiology, Kakatiya Medical College, Warangal, and were stored at 4 °C until use. The species and culture conditions of these bacterial strains are summarized in Table 2.

2.3 Blood collection and ethical compliance

Whole blood samples were collected from random healthy individuals with informed consent by the staff of Gandhi Medical College, Hyderabad, Telangana, India. The collection was conducted in accordance with the protocol approved by the Institutional Ethics Committee (IEC/GMC/2021/03/25) of the Department of Microbiology, Gandhi Medical College, Hyderabad.

2.4 Synthesis and purification of peptides

The peptides were synthesized using Fmoc (N-[9-fluorenyl]-methoxycarbonyl) chemistry, as described previously (Fields et al., 1999; Miranda et al., 1990). In this method, the amino acids were activated in situ with N-hydroxy benzotriazole (HOBt) and N, N-diisopropylethylamine (DIEA). For side-chain deprotection and peptide cleavage from the solid support, a mixture of trifluoroacetic acid (TFA), triisopropylsilane (TIS), and water was employed. Purification of the peptides was performed using preparative reverse-phase high-performance liquid chromatography (HPLC) with a Kromasil C18 column. The identity and purity of the synthesized peptides were confirmed by analytical reverse-phase HPLC and electrospray mass spectrometry.

2.5 Assessment of antibacterial activity

The minimum inhibitory concentrations (MICs) of the AMPs were determined using 96-well microtiter plates, following the guidelines of the Clinical and Laboratory Standards Institute (CLSI) (Kowalska-Krochmal & Dudek-Wicher, 2021). Briefly, each strain was inoculated into its appropriate medium (Table 2) containing varying concentrations of AMPs (100, 50, 25, 12.5, 6.25, 3.125, 1.56, and 0.75 µg/mL) in 96-well polypropylene microtiter plates. Each plate included a growth control and a negative sterile control. Bacterial inoculums were standardized to approximately 10^5 CFU/mL. Ciprofloxacin, at a concentration range of 0.5 to 1.0 µg/mL, was used as the reference antibiotic. Following incubation, the MIC was defined as the lowest concentration at which no visible growth was observed. All MICs were assessed in two independent experiments, each performed in duplicate.

2.6 Measurement of hemolytic activity (MHC)

The hemolytic activity of the peptides was evaluated by quantifying the hemoglobin released from human red blood cells (hRBCs) following peptide treatment. Freshly collected hRBCs from a healthy volunteer were drawn into polycarbonate tubes containing EDTA as an anticoagulant. The cells were washed three times with sterile phosphate-buffered saline (PBS) and centrifuged at $2,000 \times g$ for 5 min or until the supernatant became clear. The washed hRBCs were then diluted to a final concentration of 2% (vol/vol). For the assay, 50 µL of the hRBC suspension was incubated with 50 µL of peptide solutions at varying concentrations (0.98 to 250 µg/mL) prepared in PBS. Following 1 h of incubation at 37 °C, the intact hRBCs were pelleted by centrifugation at $2,000 \times g$ for 10 min. The supernatant was carefully transferred to a new 96-well plate, and the absorbance at 415 nm was measured using a PerkinElmer microplate reader to determine the amount of hemoglobin released. PBS alone (ODBlank) and 0.1% Triton X-100 (ODTriton X-100) were used as negative (0% hemolysis) and positive (100% hemolysis) controls, respectively.

The percentage of hemolysis was calculated using the formula: %Hemolysis = (ODSample – ABlank)/ (ODTritonX – 100 – ODBlack) $\times$ 100.

2.7 Animals, cell lines, and reagents

Male C57BL/6 J mice (6–8 weeks old) were sourced from the Animal Facility of Jeeva Life Sciences, Hyderabad, with original breeding colonies obtained from Jackson Labs, USA. The mice were maintained under pathogen-free conditions, and all experimental procedures were approved by the Institutional Animal Ethics Committee (IAEC) in compliance with the guidelines of the Committee for Control and Supervision of Experiments on Animals (CCSEA). Macrophage cell lines used in the study included RAW264.7 TIB-71 (mouse), BoMac (bovine), and THP-1 TIB-202 (human), initially purchased from the American Type Culture Collection (ATCC, Manassas, VA, USA). Cells were cultured in Dulbecco's Modified Eagle Medium (DMEM) or Roswell Park Memorial Institute (RPMI) 1640 medium (Sigma, USA), supplemented with 10% fetal bovine serum (FBS) (Invitrogen, Carlsbad, CA, USA), penicillin (100 U/mL), and streptomycin (100 mg/mL). The cultures were maintained at 37 °C in a humidified atmosphere with 5% CO₂. For stimulation, lipopolysaccharide (LPS) derived from *E. coli* 0111:B4 (500 ng/mL; Sigma, USA) was added to the culture medium. Peptides were introduced at concentrations of 2 µg/mL and 4 µg/mL 30 min after LPS treatment. After 24 h of incubation, the plates were centrifuged at $400 \times g$ for 6 min, and the supernatants were collected and stored at – 20 °C for cytokine analysis. Cytokine levels, including Interleukin-6 (IL-6), Interleukin-10 (IL-10), and tumor necrosis factor- α (TNF- α), were measured using species-specific sandwich ELISA kits purchased from R&D Systems.

2.8 Cellular cytotoxicity assays

Cellular cytotoxicity was evaluated using a colorimetric assay based on thiazolyl blue tetrazolium bromide (MTT; Sigma, USA). Sub-confluent monolayer cultures of RAW264.7 (mouse), BoMac (bovine), and THP-1 (human) cells were harvested using a scraper in DMEM supplemented with fetal bovine serum and antibiotics (Invitrogen, USA). The cells were seeded in 96-well microtiter plates at a density of 1.0×10^5 cells per well and exposed to various concentrations of peptide (1 to 10 µg/mL). The plates were incubated at 37 °C in a humidified atmosphere with 5% CO₂ for 48 h. After incubation, 10 µL of MTT solution (5 mg/mL) was added to each well, and the plates were further incubated for 4 h under the same conditions. Formazan crystals formed during the reaction were dissolved by adding 100 µL of 100% DMSO (Mallinckrodt Chemical, USA) to each

Table 3 Physicochemical properties of designed peptides

Peptide Id	Peptide Sequence	Support Vector Machine (SVM) classifier	Random Forest Classifier	Discriminant Analysis classifier	AMP Probability	Hydrophobic amino acid	Mol.Wt (Da)	Cationicity	total hydrophobic ratio	Protein-binding Potential (kcal/mol)	GRAVY
SK1203	AAWWKAKKWAWAKAWWKAW	1	0.9685	0.999	AMP	I:0 V:0 L:0 F:0 C:0 M:0 A:8 W:7	2659.158	6	71%	0.11	- 0.728
SK1217	AFWWKAKKFAWAKKFAWWKAW	1	0.9715	0.999	AMP	I:0 V:0 L:0 F:3 C:0 M:0 A:6 W:6	2772.318	6	71%	- 0.22	- 0.457
SK1260	KAFAVKFAWKFHAWKAWWKAW	0.994	0.9695	0.999	AMP	I:0 V:1 L:0 F:3 C:0 M:0 A:6 W:4	2636.166	6.25	53%	0.22	- 0.323
SK1286	HFAPKSKPAHFPHKSKPKAW	0.813	0.742	0.953	AMP	I:0 V:0 L:0 F:2 C:0 M:0 A:3 W:1	2454.886	6.75	29%	1.92	- 1.47
SK1281	HFAVKHAVVAHVHVKAKSR	0.502	0.6335	0.273	NAMP	I:0 V:3 L:0 F:1 C:0 M:0 A:4 W:2	2295.673	5	53%	1.4	- 0.473
SK1283	HFAVKKAKVAHFVHKAKSKAW	0.879	0.778	0.971	AMP	I:0 V:3 L:0 F:2 C:0 M:0 A:5 W:1	2418.897	6.75	52%	1.01	- 0.357

well. The plates were gently swirled for 5 min at room temperature to ensure complete dissolution of the crystals. Absorbance was measured at 575 nm using a microplate spectrophotometer (Victor X, PerkinElmer, Germany). Maximum cytotoxicity (100%) was defined by treating cells with 1% Triton X-100, while PBS served as a negative control.

2.9 Estimation of cytokine levels using enzyme-linked immunosorbent assay (ELISA)

Cultured cells or peritoneal lavage samples were centrifuged at 1000 × g for 10 min to obtain cell-free supernatants, which were then stored at −20 °C until analysis. Cytokine levels, including IL-6, IL-10, GM-CSF, MCP-1, INF-γ, and TNF-α, were measured using ELISA kits (R&D Systems) with anti-mouse cytokine-specific antibodies. The assays were performed following the manufacturer's instructions.

2.10 Structural modelling and docking studies

The amino acid sequences of six peptides (SK1203, SK1217, SK1260, SK1281, SK1283, and SK1286) were initially submitted to Blastp to search for structural homologs for homology modelling; however, no significant structural homology was identified. Consequently, the sequences were submitted for 3D structural prediction using the I-TASSER server (Roy et al., 2010). The server generated the top five predicted structures with high C-scores. From these, a structure that was consistent with the Ramachandran plot was selected for further refinement.

Energy minimization of the selected structure was performed using the steepest descent algorithm via the NOMAD-REF online program (<http://lorentz.dynstr.pasteur.fr/docking/submission.php>). The energy-minimized structure was then used for molecular docking simulations on human, bovine, and mouse TLR4 receptors using the ZDOCK server (<http://zdock.umassmed.edu>) (Pierce et al., 2014). The atomic coordinates for human, bovine, and mouse TLR4 were sourced from the crystal structures of the hTLR4 complex (PDB: 3FXI), bTLR4 (PDB: 3RG1), and mTLR4 (PDB: 3VQ1), respectively, which served as static receptor structures. The docking program generated the top 10 scoring complexes, considering factors such as shape complementarity, electrostatics, and rigid body interactions.

2.11 Protein-peptide interaction analysis using LigPlot +

The interactions between SK1260 and TLR4/MD-2 (human—3XFI, bovine—3RG1, mouse—3VQ1) were analyzed using LigPlot+ [17, 33], which generates 2D interaction maps to visualize hydrogen bonds and hydrophobic interactions. The docked protein-peptide complexes were loaded into LigPlot+, where hydrogen bonds were identified based on donor–acceptor distances (≤ 3.5 Å) and hydrophobic interactions were detected within a 4.0 Å proximity. The resulting interaction maps highlighted key residues involved in stabilizing SK1260 within the receptor binding site, allowing a comparative analysis across species.

2.12 Mouse systemic infection model

Male C57BL/6 mice, 6 weeks old and matched by weight, were used in this study. Bacterial inoculum of *E. coli* ATCC 8739 or *S. aureus* ATCC 6538 were prepared during the log phase of growth, washed, and resuspended in 0.1 mL of sterile PBS. The bacterial suspension was administered intraperitoneally (i.p.) into the mice at a dose of 5.0×10^8 CFU per mouse. Three hours after bacterial infection, mice (n = 8) were treated with either peptide SK1260 at doses of 2 mg/kg or 4 mg/kg by i.p. injection, or with an equivalent volume of PBS as a control. At 4- and 24 h post-treatment, peritoneal lavage was performed by washing the peritoneal cavity with 5 mL of sterile PBS. The lavage fluid was then collected and used for further analysis. The number of leukocytes in the lavage fluid was determined by appropriate dilution and counted using an automated cell counter. Cytokine levels in the lavage fluid, including GM-CSF, IFN-γ, and MCP-1, were measured by ELISA following the manufacturer's instructions.

2.13 Statistical analysis

For all experiments, statistical analyses were performed using Student's t-test and one-way ANOVA, where appropriate. Data are presented as the mean of triplicates ± SEM. A p-value of < 0.05 was considered statistically significant. GraphPad Prism software v9.1.0 (GraphPad Software, USA) was used for all statistical analyses.

3 Results

3.1 Peptide design and AMP prediction

Six peptides were designed based on characteristics of natural AMPs, including small size, cationicity, amphipathicity, and α -helix. The AxPEP server [3] classifies five out of the six designed peptides as antimicrobial peptides (AMPs). The peptide with the Peptide ID "SK1281" is predicted to be a Non-Antimicrobial Peptide (NAMP), and the physicochemical properties of the peptides, such as molecular weight, cationicity, hydrophobicity, protein-binding potential, and GRAVY score, provide insights into their potential functions and properties Table: 3. These predictions are based on the sequence-based classification method employed by the AxPEP server, and further experimental validation may be necessary to confirm the antimicrobial activity of these peptides.

3.2 Hemolytic activity of designed peptides on human red blood cells (RBCs)

We evaluated the hemolytic activities of six peptides (SK1203, SK1217, SK1260, SK1281, SK1283, and SK1286) using a hemolytic assay with human blood red cells (RBCs) at various concentrations. The peptides were tested at concentrations ranging from 0 $\mu\text{g/mL}$ to 100 $\mu\text{g/mL}$. We recorded the percentage of hemolysis at each peptide concentration. At all tested concentrations, the designed peptides SK1203, SK1217, SK1260, and SK1281 exhibited minimal hemolytic activity, indicating their limited ability to degrade red blood cells. At higher concentrations, SK1286 and SK1283 demonstrated noteworthy hemolytic effects. At 100 $\mu\text{g/mL}$, SK1286 caused complete hemolysis (102%), indicating a potent cytotoxic effect on RBCs. SK1283 also demonstrated concentration-dependent hemolysis, reaching 39% at 3.13 $\mu\text{g/mL}$ and 102% at 100 $\mu\text{g/mL}$. Melittin, a well-known hemolytic peptide, exhibited dose-dependent hemolytic activity, with an increasing percentage of hemolysis as the peptide concentration increased (Fig. 1).

3.3 Minimum inhibitory concentrations (MICs) of designed peptides against different microorganisms

The minimum inhibitory concentrations (MICs) of six peptides (SK1203, SK1217, SK1260, SK1281, SK1283, and SK1286) against different microorganisms are summarized in Table 5. SK1203 displayed moderate antimicrobial activity, with MIC values of 12.5 $\mu\text{g/mL}$ against *E. coli* and 25 $\mu\text{g/mL}$ against both *S. aureus* and *P. aeruginosa*. SK1217 was more effective at killing microbes. Its MIC values were 6.25 $\mu\text{g/mL}$ against *E. coli* and *K. pneumoniae*, 3.125 $\mu\text{g/mL}$ against *S. aureus*, and 12.5 $\mu\text{g/mL}$ against *P. aeruginosa*. SK1260 demonstrated potent antimicrobial activity, with MIC values of 3.13 $\mu\text{g/mL}$ against *E. coli*, *S. aureus*, *K. pneumoniae*, and *P. aeruginosa*. SK1283 showed moderate activity, with MIC values of 12.5 $\mu\text{g/mL}$ against *E. coli* and *P. aeruginosa*. SK1286 exhibited moderate to potent antimicrobial activity, with MIC values of

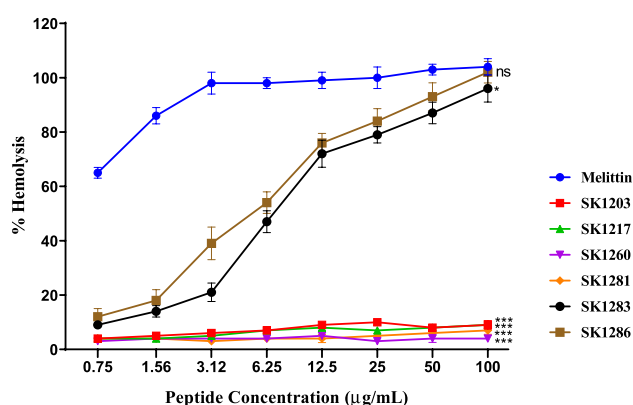


Fig. 1 Hemolytic Activity of Designed Peptides. The hemolytic assay was performed using human red blood cells (RBCs) at various peptide concentrations (0–100 $\mu\text{g/mL}$). The percentage of hemolysis was calculated relative to positive control (100% hemolysis) and negative control (0% hemolysis). Data are represented as mean $\pm$ standard deviation (SD) from three independent experiments ($n=3$). Statistical significance was assessed using one-way ANOVA followed by Dunnett's post hoc test, comparing peptide-treated groups to the control. Significance levels: $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***)

Table 4 Minimum Inhibitory concentrations of the designed peptides

	MIC (ug/mL)					
	SK1203	SK1217	SK1260	SK1281	SK1283	SK1286
<i>E.coli</i>	12.5	6.25	3.13	> 100	12.5	12.5
<i>S.aureus</i>	25	3.125	3.13	> 100	12.5	6.25
<i>K.pneumonia</i>	> 100	6.25	3.13	> 100	> 100	> 100
<i>Paeruginosa</i>	25	12.5	3.13	> 100	12.5	6.25

Table 5 Molecular docking

Peptide Id	mTLR4 (3VQ1)	hTLR4 (3FXI)	bTLR4 (3RG1)
SK1203	−281.67	−302.86	−295.04
SK1217	−301.02	−319.46	−319.43
SK1260	−327.34	−333.98	−355.69
SK1286	−315.29	−333.52	−337.04
SK1281	−262.9	−251.68	−287.25
SK1283	−268.81	−265.41	−288.75

12.5 ug/mL against *E. coli* and *P. aeruginosa*, 6.25 ug/mL against *S. aureus*, and no significant activity against *K. pneumoniae* (Suppl Fig. 1) and (Table 4). Overall, these findings indicate that the designed peptides possess varying degrees of antimicrobial activity against the tested microorganisms. SK1217 and SK1260 demonstrate stronger antimicrobial potential, while SK1203, SK1283, and SK1286 exhibit moderate activity.

3.4 Molecular docking analysis of designed peptides with TLR4/MD-2 receptor proteins

We employed molecular docking to investigate the binding patterns and affinity of the designed peptides (SK1203, SK1217, SK1260, SK1286, SK1281, and SK1283) with the receptor proteins mouse TLR4/MD-2 (3VQ1), human TLR4/MD-2 (3FXI), and bovine TLR4/MD-2 (3RG1). Table 5 summarizes the docking scores, which indicate the predicted affinity between each peptide and the respective receptor protein. Lower docking scores indicate stronger binding affinity between the peptide and the receptor protein. Based on the docking scores, SK1260 demonstrated the highest affinity for all three receptors, with docking scores of − 327.34, − 333.98, and − 355.69 for mTLR4/MD-2 (3VQ1), hTLR4/MD-2 (3FXI), and bTLR4/MD-2 (3RG1), respectively. SK1217 and SK1286 also displayed favorable binding affinities across the receptors (Fig. 2). These findings suggest that SK1260, SK1217, and SK1286 have promising potential immunomodulatory peptide candidates.

3.5 Modulation of innate immune response by cationic antimicrobial peptides in macrophage cell lines

We examined the effects of cationic antimicrobial peptides (AMPs) SK1203, SK1217, SK1260, and SK1281 on the modulation of the innate immune response in three different cell lines: murine macrophage-like RAW 264.7 cells, human macrophage cell line (THP-1), and bovine macrophage cell line (BoMac). We stimulated the cells with 500 ng/mL lipopolysaccharide (LPS) and treated them with various concentrations of the AMPs. The production of pro-inflammatory cytokines (IL-6 and TNF- α) and anti-inflammatory cytokines (IL-10) was measured. In all three cell lines, LPS stimulation led to a significant increase in IL-6 and TNF- α levels compared to the control group (PBS treatment). On the other hand, treating cells with the AMPs decreased the production of IL-6 and TNF- α while keeping IL-10 levels stable (Fig. 3A–C). Among the peptides that were tested, SK1260 had the strongest effect on stopping the production of IL-6 in RAW 264.7 and THP-1 cells. On the other hand, SK1217 had the strongest effect on stopping the production of TNF- α in BoMac cells. These results suggest that these cationic AMPs might be able to change the innate immune response by stopping the production of pro-inflammatory cytokines while keeping the production of anti-inflammatory cytokines steady in different types of cells.

Fig. 2 Binding Affinity of Designed Peptides with TLR4/MD-2 Receptors. Molecular docking analysis was performed to examine the binding patterns and affinity of six designed peptides (SK1203, SK1217, SK1260, SK1286, SK1281, and SK1283) with three receptor proteins: mouse TLR4/MD-2 (3VQ1), human TLR4/MD-2 (3FXI), and bovine TLR4/MD-2 (3RG1). The docking scores for each peptide-receptor interaction are presented below each respective image, indicating the predicted binding affinity, where lower (more negative) scores suggest stronger interactions

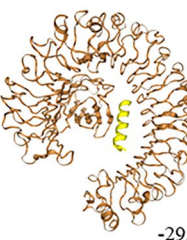
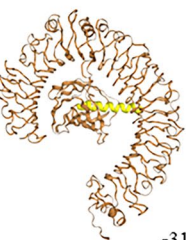
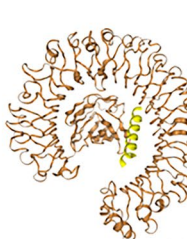
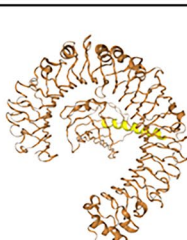
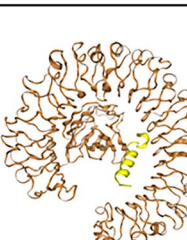
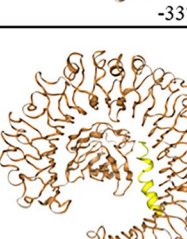
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SK1281	 -262.9	 -251.68	 -287.25
SK1283	 -268.81	 -265.41	 -288.75

Fig. 3 Illustrates the effects of cationic antimicrobial peptides (AMPs) SK1203, SK1217, SK1260, and SK1281 on the modulation of the innate immune response in different cell lines. **A** RAW 264.7 cells, **(B)** THP-1 cells, and **(C)** BoMac cells were stimulated with 500 ng/mL lipopolysaccharide (LPS) and treated with various concentrations of the AMPs. The production of pro-inflammatory cytokines IL-6 and TNF- α , as well as the anti-inflammatory cytokine IL-10, was measured

3.6 Cell viability assay of SK1260 on RAW264.7, THP-1, and BoMac cells

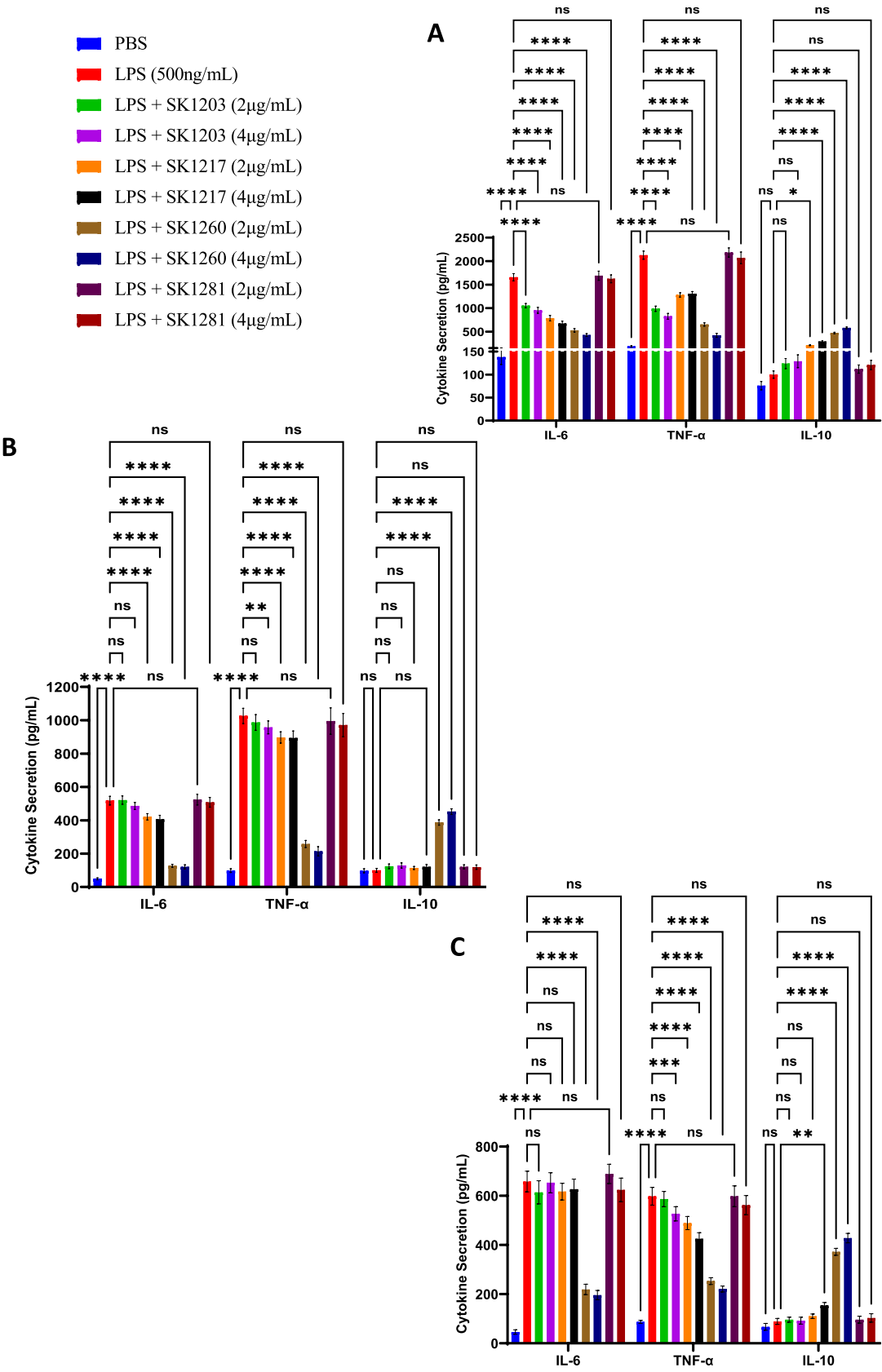
After determining the modulation of the innate immune response by peptide SK1260, we further investigated its cytotoxic effects on mouse macrophage cells RAW264.7, human macrophage cell line (THP-1), and bovine macrophage cell line (BoMac). We assessed cell viability using the MTT assay after 24 h of treatment with different concentrations of SK1260. The results showed that SK1260 exhibited minimal cytotoxicity at all tested concentrations. In RAW264.7 cells, cell viability remained high, with 99.8% at 1 μ g/mL, 97.8% at 2 μ g/mL, 98.9% at 5 μ g/mL, and 96.8% at 10 μ g/mL of SK1260. Similarly, in THP-1 cells, viability was well-preserved, with 98.2% at 1 μ g/mL, 97.6% at 2 μ g/mL, 94.8% at 5 μ g/mL, and 96.2% at 10 μ g/mL of SK1260. BoMac cells also exhibited high viability, with 99.3% at 1 μ g/mL, 96.2% at 2 μ g/mL, 97.2% at 5 μ g/mL, and 96.7% at 10 μ g/mL of SK1260 (Fig. 4). These results demonstrate that SK1260 has minimal cytotoxic effects on all three cell lines, suggesting its potential as a safe and biocompatible peptide for further therapeutic applications.

3.7 Molecular interaction analysis of SK1260 with TLR4/MD-2

The protein-peptide interaction analysis of SK1260 with TLR4/MD-2 from human (3XFI), bovine (3RG1), and mouse (3VQ1) revealed strong and stable interactions across all species. SK1260 exhibited consistent hydrogen bonding and extensive hydrophobic interactions in each complex, suggesting stable binding and potential immunomodulatory effects (Suppl Fig. 2). The interaction patterns indicate that SK1260 can effectively engage with TLR4/MD-2 in human, bovine, and mouse models, highlighting its potential cross-species applicability as an immunomodulatory peptide.

3.8 SK1260 mediated immune modulation in mice

Next, we investigated the leukocyte count in the peritoneal lavage fluid of male C57BL/6 mice infected with either *E. coli* or *S. aureus* and treated with the peptide SK1260. At 4 h post-treatment, the control group (PBS) showed a mean leukocyte count of $0.8 \times 10^6 (\pm 0.2)$ in the lavage fluid. In the *E. coli*-infected group treated with PBS, the mean leukocyte count was $0.7 \times 10^6 (\pm 0.12)$. However, treatment with SK1260 at a concentration of 2 mg/mL resulted in a significant increase in leukocyte count, with a mean of $1.2 \times 10^6 \pm 0.12$. Similarly, in the *S. aureus*-infected group, treatment with SK1260 at 2 mg/mL led to a higher leukocyte count (mean = $1.4 \times 10^6, \pm 0.12$) compared to the group treated with PBS (mean = $0.7 \times 10^6, \pm 0.18$), Fig. 5A & B. GM-CSF (Granulocyte–Macrophage Colony-Stimulating Factor), INF- γ (Interferon-gamma), and MCP-1 (Monocyte Chemoattractant Protein-1) are key cytokines involved in the immune response and inflammation [7], Gil- [9, 11, 29]. Analyzing these cytokines provides valuable insights into the immune modulation and recruitment of immune cells in response to bacterial infection. We next checked the levels of these cytokines (GM-CSF, INF- γ , and MCP-1) in the peritoneal lavage fluid. At 4 h post-treatment, the control group (PBS) exhibited basal levels of cytokines. In the *E. coli*-infected group treated with SK1260 (2 mg/mL), there was an elevation in GM-CSF (mean = $627, \pm 38$), INF- γ (mean = $534, \pm 35$), and MCP-1 (mean = $728, \pm 43$) compared to the PBS-treated group (GM-CSF mean = 324, INF- γ mean = 128, MCP-1 mean = 126). Similarly, in the *S. aureus*-infected group treated with SK1260 (2 mg/mL), higher levels of GM-CSF (mean = $572, \pm 41$), INF- γ (mean = $434, \pm 41$), and MCP-1 (mean = $628, \pm 51$) were observed compared to the PBS-treated group (GM-CSF mean = 243, INF- γ mean = 218, MCP-1 mean = 216), Fig. 6A & B. These findings demonstrate that treatment with SK1260 can modulate leukocyte recruitment and cytokine production in response to bacterial infection. The increased leukocyte count suggests enhanced immune cell response, while the elevated levels of GM-CSF, INF- γ , and MCP-1 indicate amplified immune activation and recruitment of immune cells to the site of infection. These results offer significant insights into the immune modulation SK1260 mediates, underscoring its potential as a therapeutic agent for immune modulation in infectious diseases.



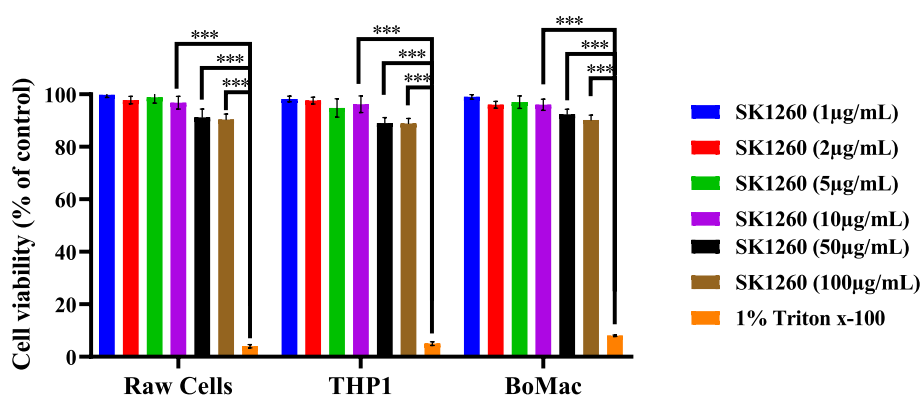


Fig. 4 Cell Viability Assay of Peptide SK1260. Cell viability was assessed using the MTT assay after 24 h of treatment with varying concentrations of SK1260 in RAW264.7 cells, THP-1 cells, and BoMac cells. The results indicate that SK1260 exhibited minimal cytotoxicity across all tested cell lines. Data are presented as mean $\pm$ standard deviation (SD) from three independent experiments ($n=3$), with statistical significance analyzed using one-way ANOVA followed by Dunnett's post hoc test. Significance levels: $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***).

4 Discussion

Antimicrobial peptides (AMPs) are renowned for their antimicrobial activity, a trait attributed to distinct characteristics such as small size, cationicity, amphipathicity, and α -helical structure [1], R. E. W. [12]. Inspired by these features, our research focused on designing, characterizing, and evaluating six peptides to endow them with antimicrobial and immunomodulatory properties. Notably, tools like the AxPEP server identified five out of these six peptides as potential AMPs, suggesting broad-spectrum antimicrobial activity against various pathogens.

Hemolytic activity is a critical safety parameter in evaluating peptides for therapeutic applications. Hemolysis assays, which assess a peptide's propensity to lyse red blood cells, are crucial for understanding the safety profile of AMPs [22, 28]. Our findings were encouraging, as four of our peptides, including SK1203, SK1217, SK1260, and SK1281, demonstrated minimal hemolytic activity across all tested concentrations.

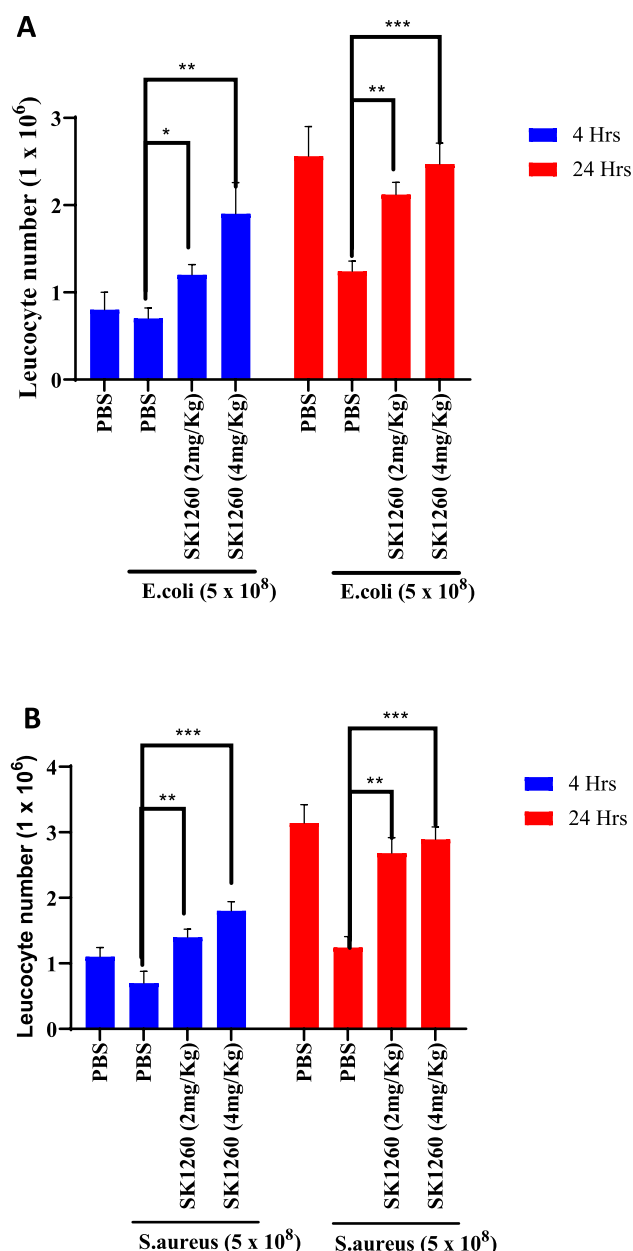
We further scrutinized antimicrobial activity using Minimum Inhibitory Concentrations (MICs) against diverse bacterial strains. Two peptides, SK1217 and SK1260, exhibited significant antimicrobial activity, underscoring their potential as broad-spectrum antimicrobial agents. Molecular docking studies, a valuable tool for predicting peptide interactions with receptor proteins [20, 31], revealed SK1260's pronounced affinity for several receptor proteins. This suggests a potential immunomodulatory role for SK1260, implying that it might modulate immune responses mediated by these receptors.

We assessed the innate immune response modulation of these peptides using macrophage cell lines. After treatment with the AMPs, we observed a concentration-dependent reduction in the production of pro-inflammatory cytokines IL-6 and TNF- α . Interestingly, production levels of the anti-inflammatory cytokine IL-10 remained stable. Such selective targeting of pro-inflammatory cytokines is desirable, as imbalances in cytokine production can exacerbate inflammatory diseases [25]. In this context, our peptides, especially SK1260, could play a pivotal role in regulating excessive inflammatory responses.

Further cytotoxicity studies on SK1260 confirmed minimal cytotoxic effects on macrophage cell lines, reinforcing its therapeutic safety potential. In vivo studies complemented these findings; treatment with SK1260 resulted in enhanced leukocyte recruitment in mice infected with *E. coli* or *S. aureus*. Elevated cytokine levels in SK1260-treated groups, such as GM-CSF, IFN- γ , and MCP-1, point to its role in modulating cytokine production during bacterial infections.

Several AMPs, such as LL-37, derived from human cathelicidin and human β -defensins, showcase dual antimicrobial and immunomodulatory functions [21]. Their multifaceted modes of action, which include direct microbial killing and immune response modulation, inspired our peptide designs. Minor sequence adjustments, influenced by the type and position of amino acid residues, can profoundly impact a peptide's antimicrobial activity and interaction with microbial membranes [8]. In our study, these sequence variations led to distinct outcomes in antimicrobial efficacy, hemolytic profiles, and immunomodulatory potential. Among these, SK1260 stood out, largely due to its unique sequence and amphipathic nature. Challenges such as protease degradation and potential cytotoxicity persist, highlighting the need for continuous refinement in AMP development [19]. Additionally, this study lacks TEM analysis, which could provide direct visualization of bacterial membrane disruption and the antimicrobial activity was tested against a limited set of bacterial strains, leaving its

Fig. 5 Leukocyte Counts in Peritoneal Lavage Fluid. Leukocyte counts in the peritoneal lavage fluid of mice infected with (A) *E. coli* or (B) *S. aureus* and treated with different concentrations of SK1260 or PBS. Data are expressed as mean $\pm$ standard deviation (SD) from at least three independent experiments ($n=3-5$ per group). Statistical significance was analyzed using one-way ANOVA followed by Dunnett's post hoc test. Significance levels: $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***)

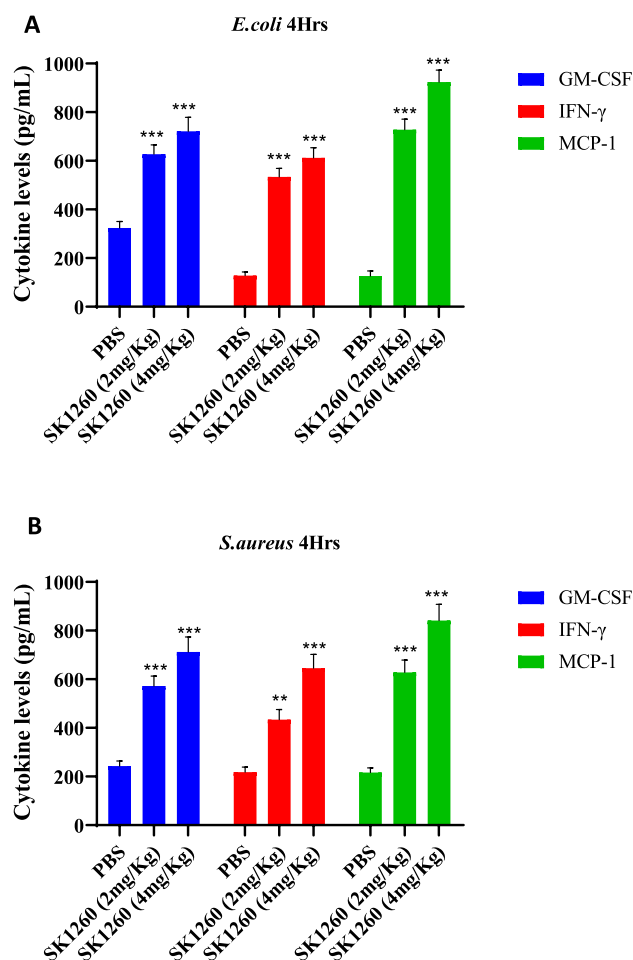


efficacy against multidrug-resistant bacteria and fungi unexplored. Future studies will address these limitations to further validate the therapeutic potential of SK1260.

5 Conclusion

In conclusion, our study focused on the design and characterization of cationic AMPs with dual antimicrobial and immunomodulatory activities. The designed peptides had different levels of antimicrobial activity against Gram-negative and Gram-positive bacteria. SK1217 and SK1260 were the most effective against a wide range of bacteria. The peptides demonstrated minimal hemolytic activity, indicating a low potential for toxicity. Molecular docking studies showed that they might be able to interact with receptor proteins that help control the immune response. This shows that

Fig. 6 Cytokine Levels in Peritoneal Lavage Fluid. Cytokine levels (GM-CSF, IFN- γ , and MCP-1) in the peritoneal lavage fluid of mice infected with (A) *E. coli* or (B) *S. aureus* and treated with different concentrations of SK1260 or PBS. Data represent mean $\pm$ standard deviation (SD) from three independent experiments. Statistical analysis was performed using one-way ANOVA followed by Dunnett's post hoc test. Significance levels: $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***)



they have immunomodulatory properties. The AMPs effectively modulated cytokine production in macrophage cell lines, selectively suppressing pro-inflammatory cytokines while maintaining anti-inflammatory cytokine levels. SK1260 exhibited minimal cytotoxicity and showed promising effects in promoting immune cell recruitment and cytokine production in infected mice. These findings emphasize the potential of cationic AMPs as therapeutic agents with dual antimicrobial and immunomodulatory properties. We warrant further investigations, including in vivo efficacy studies and toxicity assessments, to fully explore their clinical applications.

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Author contributions KS, MKT, and SC conceptualized the study and designed the experiments. SK, KS, and MK carried out the experiments. Data analysis was performed by SK, KS, MK, and SC. KS prepared all the figures and conducted the statistical analysis. SK and KS drafted the initial manuscript, while MKT and SC provided critical revisions and edits. All authors reviewed and approved the final version of the manuscript.

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Data availability All data generated or analyzed during this study are included in this published article and its supplementary information files. Further information is available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate All animal experiments were approved by the Institutional Animal Ethics Committee of Jeeva Life Sciences, Hyderabad and were conducted in accordance with relevant guidelines and regulations. Human studies were approved by the

Department of Microbiology, Gandhi Medical College, Secunderabad, and conducted in accordance with institutional and national guidelines. Informed written consent was obtained from all participants prior to their inclusion in the study.

Competing interests The authors declare no competing interests.

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